

Finding the best linear approximation of a set of points on a Cartesian plane

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1 Problem

Given a set of N points specified using their coordinates $P_i = (x_i, y_i)$ on a Cartesian plane, with $i \in [1, N]$, find the best linear function (i.e. a straight line) that minimizes a certain cost function f .

The expression of the straight line on a Cartesian plane is:

$$y = mx + q$$

where m and q are the unknowns. The goal is exactly to find the value of these two parameters to minimize a given cost function. The cost function can be chosen depending on the actual problem to be solved.

1.1 Cost functions

One of the most common cost functions is the *Mean Square Error* (MSE). For each point P_i , the error is defined as the difference between its ordinate y_i and the ordinate of the linear approximation at the corresponding abscissa x_i ; therefore the error and the square of the error for point P_i are:

$$\varepsilon_i = y_i - (mx_i + q) \quad \varepsilon_i^2 = (y_i - (mx_i + q))^2$$

The Mean Square Error is simply the average of the ε_i^2 , obtained by summing all the square errors ε_i^2 together and dividing the result by the number of points N .

$$f(m, q) = \varepsilon_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \varepsilon_i^2 = \frac{1}{N} \sum_{i=1}^N (y_i - (mx_i + q))^2$$

To solve the stated problem, the coordinates (m, q) of the minimum of this cost function should be found. This technique to determine the best linear approximation is known as *Minimum Mean Square Error* (MMSE), but also as *Least Mean Square* or simply *Least Squares*.

More rarely, the Mean Absolute Error is used. While for the MMSE an analytical solution is found in the following, this is not possible for minimizing the Mean Absolute Error; however, this cost function might be more efficiently implemented in numerical systems compared to the MMSE, as it doesn't need to take the squares. The absolute error for point P_i is defined as:

$$|\varepsilon_i| = |y_i - (mx_i + q)|$$

and the Mean Absolute Error is computed by summing all errors $|\varepsilon_i|$ and dividing the result by the number of points N .

$$f(m, q) = \varepsilon_{\text{MAE}} = \frac{1}{N} \sum_{i=1}^N |\varepsilon_i| = \frac{1}{N} \sum_{i=1}^N |y_i - (mx_i + q)|$$

Again, to solve the stated problem, the coordinates (m, q) of the minimum of this cost function should be found. Note that while x_i and y_i are known values, m and q , which appear in the expression of the errors, are not. Therefore, regardless of the chosen method, the cost function is a function $f(m, q) : \mathbb{R}^2 \rightarrow \mathbb{R}$, i.e. it is a surface in a three dimensional space for which we should find the minimum¹.

¹In more general cases the cost function is $f : \mathbb{R}^n \rightarrow \mathbb{R}$, but in this description it is limited to just $n = 2$, i.e. just two dimensions.

1.2 Example

Assume you have $N = 4$ points with the following coordinates:

$$P_1 = (0, 0.1) \quad P_2 = (1, 0.55) \quad P_3 = (2, 0.95) \quad P_4 = (3, 1.6)$$

They are represented in Figure 1 on the x, y Cartesian plane. They are roughly lying on the straight line with equation:

$$y = \frac{1}{2}x$$

This line has $m = 0.5$ and $q = 0$ and it is *not* the best linear approximation of the 4 points. In the following, the best linear approximation is calculated using the MMSE technique, and the result is compared to the expression of this straight line.

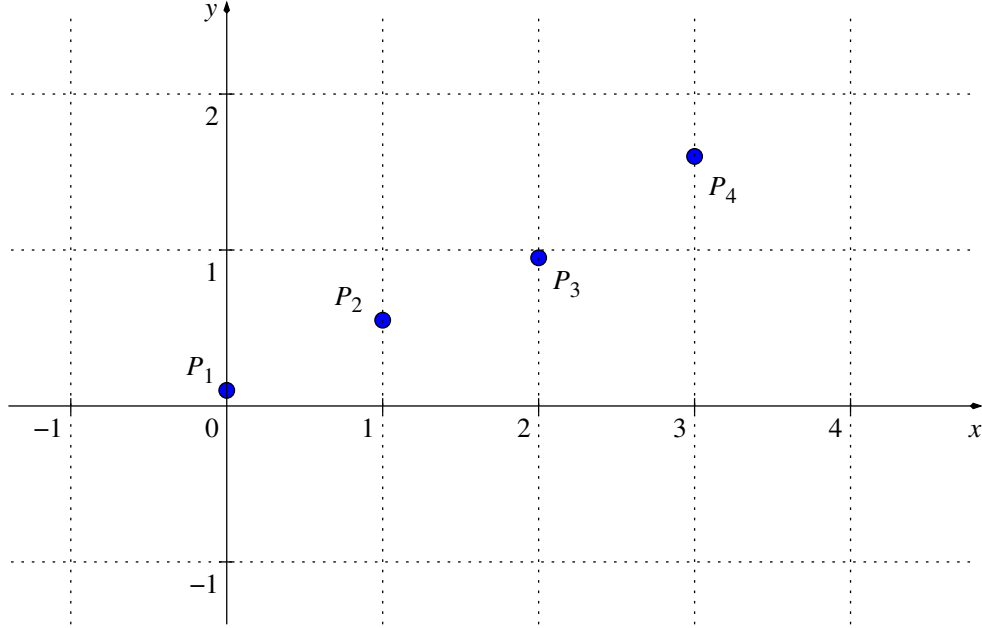


Figure 1: Example points on the x, y Cartesian plane

2 Solution for MMSE

Let's write the general expression of the Mean Square Error ϵ_{MSE} :

$$\begin{aligned} \epsilon_{\text{MSE}} &= \frac{1}{N} \sum_{i=1}^N \epsilon_i^2 = \frac{1}{N} \sum_{i=1}^N (y_i - (mx_i + q))^2 = \frac{1}{N} \sum_{i=1}^N (y_i^2 + m^2 x_i^2 + q^2 - 2mx_i y_i - 2qy_i + 2mx_i q) = \\ &= \frac{1}{N} \sum_{i=1}^N y_i^2 + \frac{1}{N} \left(\sum_{i=1}^N x_i^2 \right) m^2 + \frac{1}{N} \left(\sum_{i=1}^N 1 \right) q^2 - 2 \frac{1}{N} \left(\sum_{i=1}^N x_i y_i \right) m - 2 \frac{1}{N} \left(\sum_{i=1}^N y_i \right) q + 2 \frac{1}{N} \left(\sum_{i=1}^N x_i \right) m q = \\ &= \frac{1}{N} \sum_{i=1}^N y_i^2 + \frac{1}{N} \left(\sum_{i=1}^N x_i^2 \right) m^2 + \frac{1}{N} N q^2 - 2 \frac{1}{N} \left(\sum_{i=1}^N x_i y_i \right) m - 2 \frac{1}{N} \left(\sum_{i=1}^N y_i \right) q + 2 \frac{1}{N} \left(\sum_{i=1}^N x_i \right) m q = \\ &= \frac{1}{N} \left(\sum_{i=1}^N x_i^2 \right) m^2 + q^2 + \frac{2}{N} \left(\sum_{i=1}^N x_i \right) m q - \frac{2}{N} \left(\sum_{i=1}^N x_i y_i \right) m - \frac{2}{N} \left(\sum_{i=1}^N y_i \right) q + \frac{1}{N} \sum_{i=1}^N y_i^2 = \\ &= \frac{A}{N} m^2 + q^2 + \frac{2B}{N} m q - \frac{2C}{N} m - \frac{2D}{N} q + \frac{E}{N} \end{aligned}$$

where

$$A = \sum_{i=1}^N x_i^2, \quad B = \sum_{i=1}^N x_i, \quad C = \sum_{i=1}^N x_i y_i, \quad D = \sum_{i=1}^N y_i, \quad E = \sum_{i=1}^N y_i^2$$

All parameters A , B , C , D and E are known since they only depend on the points coordinates. The expression for ε_{MSE} is a quadratic surface, and in particular it is a positive paraboloid in a space where m and q are the independent variables and ε_{MSE} is the dependent one. Since both squared terms, m^2 and q^2 , have a positive coefficient, the paraboloid opens upwards (it holds water), and the vertex corresponds to its minimum. The coordinates of the vertex in the (m, q) plane are exactly the solution we are looking for, as the MSE is minimum in that point, and the vertex height is the value of the minimum MSE.

Given the above discussion, it is sufficient to set the partial derivative of ε_{MSE} with respect to m and q to zero (set the gradient to zero) to find the minimum (a maximum or a saddle point would also have a zero gradient, but the surface is guaranteed not to have them, so further checks are not required). Therefore:

$$\begin{cases} \frac{\partial \varepsilon_{\text{MSE}}}{\partial m} = \frac{2A}{N}m + \frac{2B}{N}q - \frac{2C}{N} = 0 \\ \frac{\partial \varepsilon_{\text{MSE}}}{\partial q} = \frac{2B}{N}m + 2q - \frac{2D}{N} = 0 \end{cases} \quad \begin{cases} Am + Bq = C \\ Bm + Nq = D \end{cases} \quad \begin{cases} m = \frac{NC - BD}{NA - B^2} = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2} \\ q = \frac{AD - BC}{NA - B^2} = \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{N \sum x_i^2 - (\sum x_i)^2} \end{cases}$$

The expressions found for m and q depend only on known values, so the problem is solved in a closed form and the linear approximation that minimizes the Mean Square Error is determined. Any other combination of values for m and q gives a straight line whose MSE is higher.

2.1 Notable cases

2.1.1 Points are on a vertical line

The above expressions are valid only when the denominator $(NA - B^2)$ is different from zero. The denominator is zero when all x_i are identical, i.e. $x_i = X, \forall i \in [1, N]$. In this case the denominator becomes:

$$NA - B^2 = N \sum_{i=1}^N x_i^2 - \left(\sum_{i=1}^N x_i \right)^2 = N \sum_{i=1}^N X^2 - \left(\sum_{i=1}^N X \right)^2 = NX^2 \left(\sum_{i=1}^N 1 \right) - X^2 \left(\sum_{i=1}^N 1 \right)^2 = NX^2 - NX^2 = 0$$

Assuming y_i are different, this case corresponds to having all points P_i along a vertical line, and the best linear approximation would be the expression $x = X$, which is not in the form $y = mx + q$ assumed initially.

2.1.2 Points are on an horizontal line

Note also that if all y_i are identical, i.e. $y_i = Y, \forall i \in [1, N]$, and the x_i are different (the denominator is therefore non zero), then the numerator for m becomes 0, and therefore $m = 0$, and we get $q = Y$, as detailed below:

$$\begin{aligned} NC - BD &= N \sum_{i=1}^N x_i y_i - \sum_{i=1}^N x_i \sum_{i=1}^N y_i = N \sum_{i=1}^N x_i Y - \sum_{i=1}^N x_i \sum_{i=1}^N Y = NY \sum_{i=1}^N x_i - Y \sum_{i=1}^N x_i \sum_{i=1}^N 1 = \\ &= NY \sum_{i=1}^N x_i - NY \sum_{i=1}^N x_i = 0 \\ m &= \frac{0}{NA - B^2} = 0 \\ AD - BC &= \sum_{i=1}^N x_i^2 \sum_{i=1}^N y_i - \sum_{i=1}^N x_i \sum_{i=1}^N x_i y_i = \sum_{i=1}^N x_i^2 \sum_{i=1}^N Y - \sum_{i=1}^N x_i \sum_{i=1}^N x_i Y = Y \sum_{i=1}^N x_i^2 \sum_{i=1}^N 1 - Y \sum_{i=1}^N x_i \sum_{i=1}^N x_i = \\ &= NY \sum_{i=1}^N x_i^2 - Y \left(\sum_{i=1}^N x_i \right)^2 = Y \left[N \sum_{i=1}^N x_i^2 - \left(\sum_{i=1}^N x_i \right)^2 \right] = Y (NA - B^2) \\ q &= \frac{Y (NA - B^2)}{NA - B^2} = Y \end{aligned}$$

which are the expected results, as all points lie on an horizontal line. The best linear approximation perfectly matches all points, so the Minimum MSE should be zero; in fact:

$$\begin{aligned}\varepsilon_{\text{MSE}}(0, Y) &= \frac{A}{N} \cdot 0^2 + Y^2 + \frac{2B}{N} \cdot 0 \cdot Y - \frac{2C}{N} \cdot 0 - \frac{2D}{N} Y + \frac{E}{N} = Y^2 - \frac{2D}{N} Y + \frac{E}{N} = Y^2 - \frac{2Y}{N} \sum_{i=1}^N y_i + \frac{1}{N} \sum_{i=1}^N y_i^2 = \\ &= Y^2 - \frac{2Y}{N} \sum_{i=1}^N Y + \frac{1}{N} \sum_{i=1}^N Y^2 = Y^2 - \frac{2Y}{N} Y \sum_{i=1}^N 1 + \frac{1}{N} Y^2 \sum_{i=1}^N 1 = Y^2 - \frac{2Y}{N} Y N + \frac{1}{N} Y^2 N = Y^2 - 2Y^2 + Y^2 = 0\end{aligned}$$

2.1.3 Points are coincident

If, on the other hand, all x_i are identical, i.e. $x_i = X, \forall i \in [1, N]$, and all y_i are identical, i.e. $y_i = Y, \forall i \in [1, N]$, the numerator and the denominator for both m and q become zero, and the result is not determined. This case corresponds to having all points coincident. It is still possible to find solutions to this problem, as any straight line passing through the point of coordinates $P = (X, Y)$ gives $\varepsilon_{\text{MSE}} = 0$, and the general equation for these lines is:

$$y - Y = m(x - X)$$

where m can be any value (with $m = \infty$ then the line is $x = X$, i.e. a vertical line), and $q = Y - mX$, as also proved in the next paragraph. Any straight line not passing through P gives an MSE which is higher than zero, so it is not minimum.

2.1.4 Statistical interpretation

Additionally consider that the expressions for m and q can be rewritten based on statistical quantities:

$$\begin{aligned}NA - B^2 &= N \sum_{i=1}^N x_i^2 - \left(\sum_{i=1}^N x_i \right)^2 = N^2 \frac{\sum_{i=1}^N x_i^2}{N} - \left(N \frac{\sum_{i=1}^N x_i}{N} \right)^2 = N^2 E(x_i^2) - N^2 E(x_i)^2 = N^2 \sigma_{x_i} \\ NC - BD &= N \sum_{i=1}^N x_i y_i - \sum_{i=1}^N x_i \sum_{i=1}^N y_i = N^2 \frac{\sum_{i=1}^N x_i y_i}{N} - N \frac{\sum_{i=1}^N x_i}{N} N \frac{\sum_{i=1}^N y_i}{N} = N^2 E(x_i y_i) - N^2 E(x_i) E(y_i) = \\ &= N^2 (E(x_i y_i) - E(x_i) E(y_i)) = N^2 \sigma_{x_i y_i} \\ m &= \frac{NC - BD}{NA - B^2} = \frac{N^2 \sigma_{x_i y_i}}{N^2 \sigma_{x_i}} = \frac{\sigma_{x_i y_i}}{\sigma_{x_i}} \\ q &= \frac{D - Bm}{N} = \frac{D}{N} - \frac{B}{N} m = \frac{\sum_{i=1}^N y_i}{N} - \frac{\sum_{i=1}^N x_i}{N} m = E(y_i) - \frac{\sigma_{x_i y_i}}{\sigma_{x_i}} E(x_i)\end{aligned}$$

where $E(\cdot)$ is the expected value (mean, average), and σ is the variance or co-variance. So the slope of the best linear approximation is the ratio between the covariance of x_i and y_i , and the variance of x_i , while q is related to the averages of x_i and y_i and the slope. The best linear approximation is in fact the straight line with the slope obtained by the ratio of the covariance with the variance, and passing through the point $\bar{P} = (E(x_i), E(y_i))$, i.e. the point with abscissa the average x_i and ordinate the average y_i .

2.1.5 Two points in total

As a last point consider the case when $N = 2$, $P_1 = (X_1, Y_1)$ and $P_2 = (X_2, Y_2)$. In this situation the best linear approximation is the straight line passing through the two points. This is indeed what is obtained by substituting the values in the above expressions:

$$A = \sum_{i=1}^N x_i^2 = X_1^2 + X_2^2, \quad B = \sum_{i=1}^N x_i = X_1 + X_2, \quad C = \sum_{i=1}^N x_i y_i = X_1 Y_1 + X_2 Y_2, \quad D = \sum_{i=1}^N y_i = Y_1 + Y_2, \quad E = \sum_{i=1}^N y_i^2 = Y_1^2 + Y_2^2$$

$$NC - BD = 2(X_1 Y_1 + X_2 Y_2) - (X_1 + X_2)(Y_1 + Y_2) = (X_1 - X_2)(Y_1 - Y_2)$$

$$AD - BC = (X_1^2 + X_2^2)(Y_1 + Y_2) - (X_1 + X_2)(X_1 Y_1 + X_2 Y_2) = (X_1 - X_2)(X_1 Y_2 - X_2 Y_1)$$

$$\begin{aligned}
NA - B^2 &= 2(X_1^2 + X_2^2) - (X_1 + X_2)^2 = (X_1 - X_2)^2 \\
m &= \frac{NC - BD}{NA - B^2} = \frac{(X_1 - X_2)(Y_1 - Y_2)}{(X_1 - X_2)^2} = \frac{Y_1 - Y_2}{X_1 - X_2} \\
q &= \frac{AD - BC}{NA - B^2} = \frac{(X_1 - X_2)(X_1 Y_2 - X_2 Y_1)}{(X_1 - X_2)^2} = \frac{X_1 Y_2 - X_2 Y_1}{X_1 - X_2}
\end{aligned}$$

which are the expected results. Since the best linear approximation passes through the two points P_1 and P_2 in this case, the minimum ε_{MSE} should be zero. Try substituting the obtained values for A, B, C, D, E, N, m and q in the expression for ε_{MSE} to verify this claim (this is a long calculation and it is not reported here).

2.2 Example (continued)

Using the 4 points previously defined, the expression for the MSE is:

$$\varepsilon_{\text{MSE}} = \frac{1}{4} [(0.1 - (m \cdot 0 + q))^2 + (0.55 - (m \cdot 1 + q))^2 + (0.95 - (m \cdot 2 + q))^2 + (1.6 - (m \cdot 3 + q))^2]$$

Also, the values for paramters N, A, B, C, D , and E can be computed. They are respectively:

$$N = 4, \quad A = 14, \quad B = 6, \quad C = 7.25, \quad D = 3.2, \quad E = 3.775$$

Either by carrying out the counts, or by substituting the parameters in the expression derived for the MSE solution, we have:

$$\varepsilon_{\text{MSE}} = 3.5m^2 + q^2 + 3mq - 3.625m - 1.6q + 0.94375$$

As stated above, this is a paraboloid, as can be seen in Figure 2, and the coordinates of its minimum should be determined. In the figure the names of the axes are not given, but the m axis goes horizontally from -1.5 to 2.5 , the q axis goes from -2.5 to 1.5 , while ε_{MSE} is vertical from 0 to 5 . From the contour lines on the base of the graph in Figure 2 it is clear that the solution is near $m = 0.5$ and $q = 0$, but the exact solution is actually slightly different.

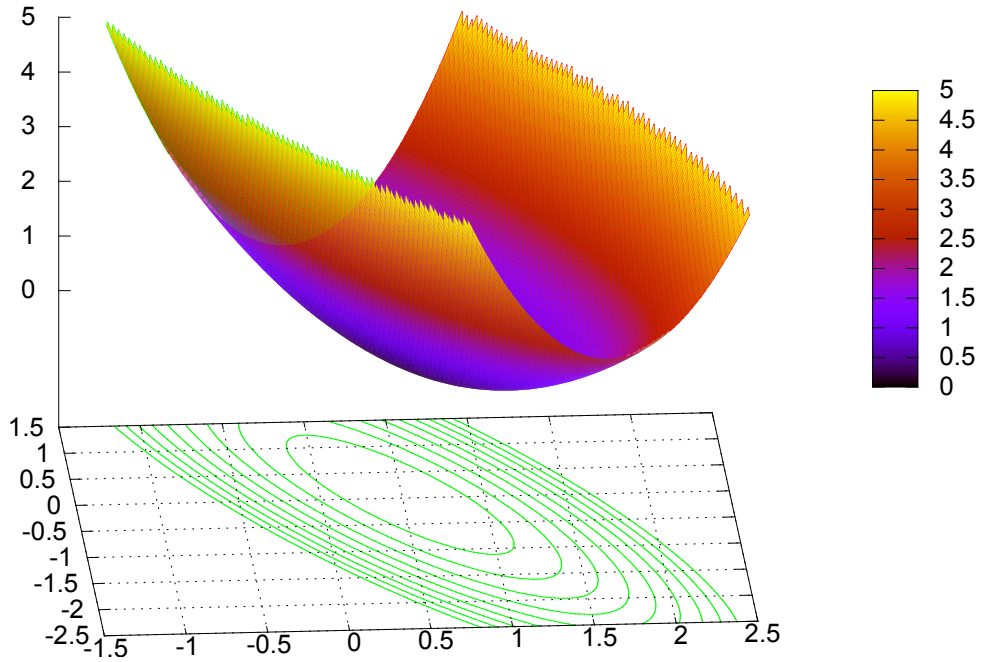


Figure 2: MSE paraboloid

Using the closed form solution, the coordinates of the minimum are given by:

$$\begin{cases} m = \frac{4 \cdot 7.25 - 6 \cdot 3.2}{4 \cdot 14 - 36} = 0.49 \\ q = \frac{14 \cdot 3.2 - 6 \cdot 7.25}{4 \cdot 14 - 36} = 0.065 \end{cases}$$

The minimum value for ε_{MSE} is:

$$\varepsilon_{\text{MSE}}^{\min} = 3.625 \cdot 10^{-3}$$

The values obtained for m and q are very close to $1/2$ and 0 , which were those roughly obtained by eye inspection. Note that with $m = 0.5$ and $q = 0$ the MSE is $\varepsilon = 6.25 \cdot 10^{-3}$, which is clearly higher than the minimum one. Going further away from the best solution the MSE rapidly increases, as can be clearly seen in Figure 2. Figure 3 shows the best linear approximation along with the four points. Note moreover that $E(x_i) = 1.5$ and $E(y_i) = 0.8$, and indeed the best linear approximation passes through the point $\bar{P} = (1.5, 0.8)$.

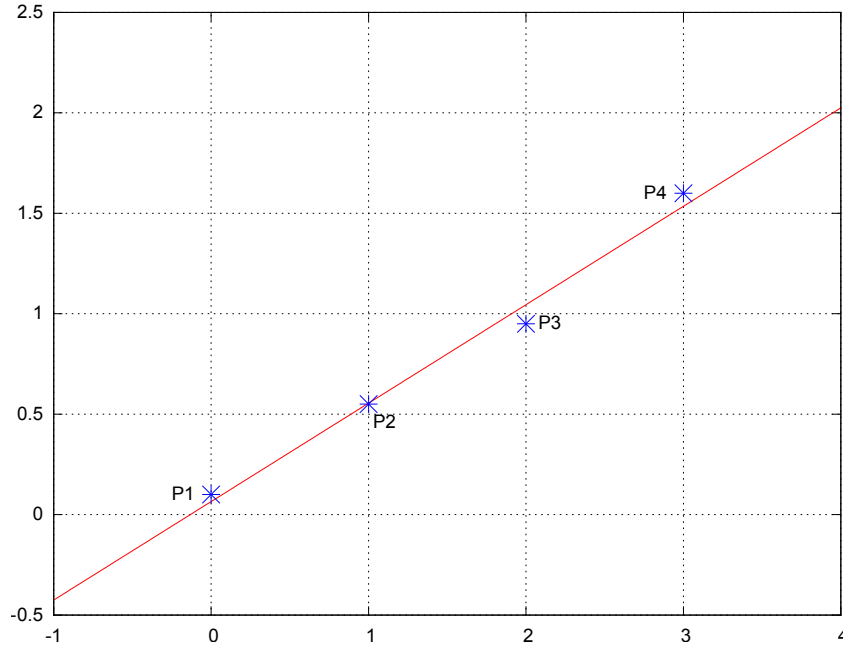


Figure 3: Best linear approximation using MSE

2.2.1 Mean Absolute Error example

For your reference, Figure 4 shows the Mean Absolute Error surface with contour lines on the base of the graph. Note that, although the minimum is extremely close to the MSE case, the solution is actually different (the real values are not computed). Compared to the MSE paraboloid, the surface grows less rapidly when going away from the best solution. From these two examples it can be said that, in general, changing the cost function gives a different best linear approximation.

3 Best linear approximation using a calculator or PC

Instead of performing all calculations manually, you may wish to use some dedicated tools. Many pocket scientific calculators include statistical functions such as the *Linear Regression*, that give the parameters m and q once all the N points are introduced. See the instruction manual for the specific calculator for more details.

The PC program Microsoft Excel also includes linear regression among its tools. It is found under *Data Analysis* toolset (which may not be installed by default) and gives a lot of statistical information. The data should be organized as two columns, one for the x_i and the other for the corresponding y_i . The *LINEST* function also provides similar information. Otherwise just plot an *XY Scatter* graph of the data and activate the *Trendline*; ensure to select the linear approximation, and select to display the equation on the chart. Figure 5 shows the example in Excel, with both the result obtained using the Data Analysis tool and the graph with the trendline.

3.1 Best linear approximation in Matlab/Octave

Matlab and Octave are very powerful tools to perform operations on vectors and matrices. It turns out that the linear regression problem can be effectively formulated in a vectorized fashion, that leads to an easy implementation using these

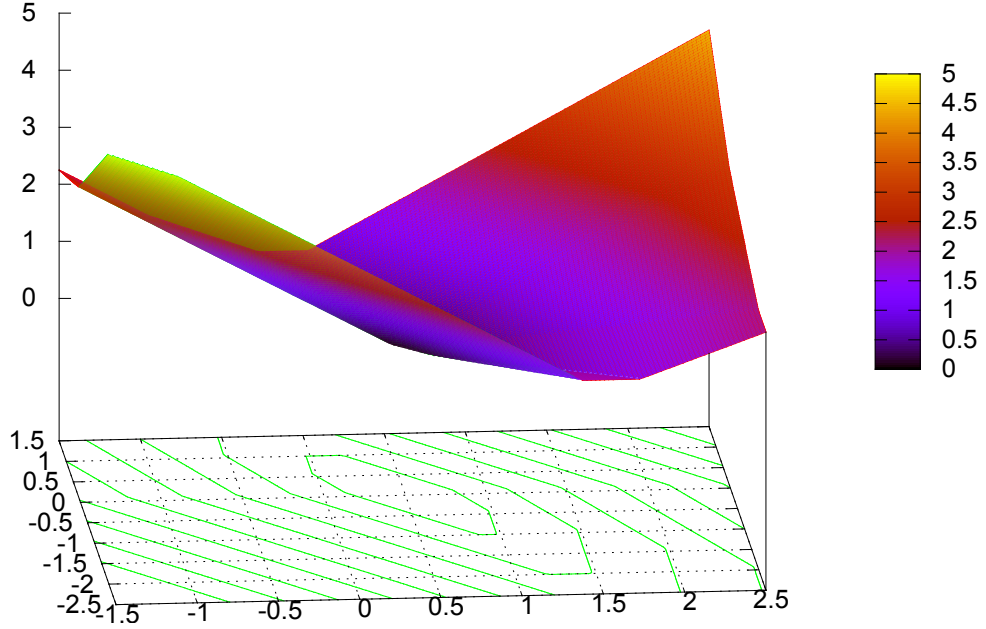


Figure 4: Mean Absolute Error surface

tools.

Suppose we have N points, each specified by its coordinates $P_i = (x_i, y_i)$. Let's construct an $N \times 2$ matrix X that has N rows and 2 columns, where the first column is made of 1s, and the second column by the values x_i ; let's construct a column vector y of N elements, where each element is y_i , and a column vector θ of 2 elements, where elements are q and m . The following is an example:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_N \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_N \end{bmatrix} \quad \theta = \begin{bmatrix} q \\ m \end{bmatrix}$$

Knowing that given a generic column vector v , the sum of the squares of all its elements can be computed as $v^T v$, where the superscript T means transposition (of a vector or a matrix), then the cost function for the Mean Square Error ε_{MSE} can be simply written in a matrix form as in the following expression:

$$\varepsilon_{\text{MSE}} = \frac{1}{N} (y - X\theta)^T (y - X\theta)$$

In fact the quantity $(y - X\theta)$ is a column vector of N elements, where each element is the error ε_i computed for point P_i as:

$$\varepsilon_i = y_i - 1 \cdot q - x_i \cdot m$$

and we are taking the sum of the squares by performing the vector multiplication. The content of the vector θ are the two parameters m and q , that are the unknowns. We want to find the values of m and q , i.e. the elements of θ , to minimize ε_{MSE} . To do so let's expand ε_{MSE} and then take the derivatives (the gradient) and set them to zero:

$$\varepsilon_{\text{MSE}} = \frac{1}{N} (y^T y - (X\theta)^T y - y^T (X\theta) + (X\theta)^T (X\theta)) = \frac{1}{N} (y^T y - 2\theta^T X^T y + \theta^T X^T X\theta)$$

This result is obtained knowing that $y^T (X\theta) = (X\theta)^T y$ and $(X\theta)^T = \theta^T X^T$ (due to properties of matrix multiplication and transposition). The partial derivatives with respect to θ then are:

$$\frac{\partial \varepsilon_{\text{MSE}}}{\partial \theta} = \frac{1}{N} (-2X^T y + 2X^T X\theta) = 0 \implies X^T X\theta = X^T y$$

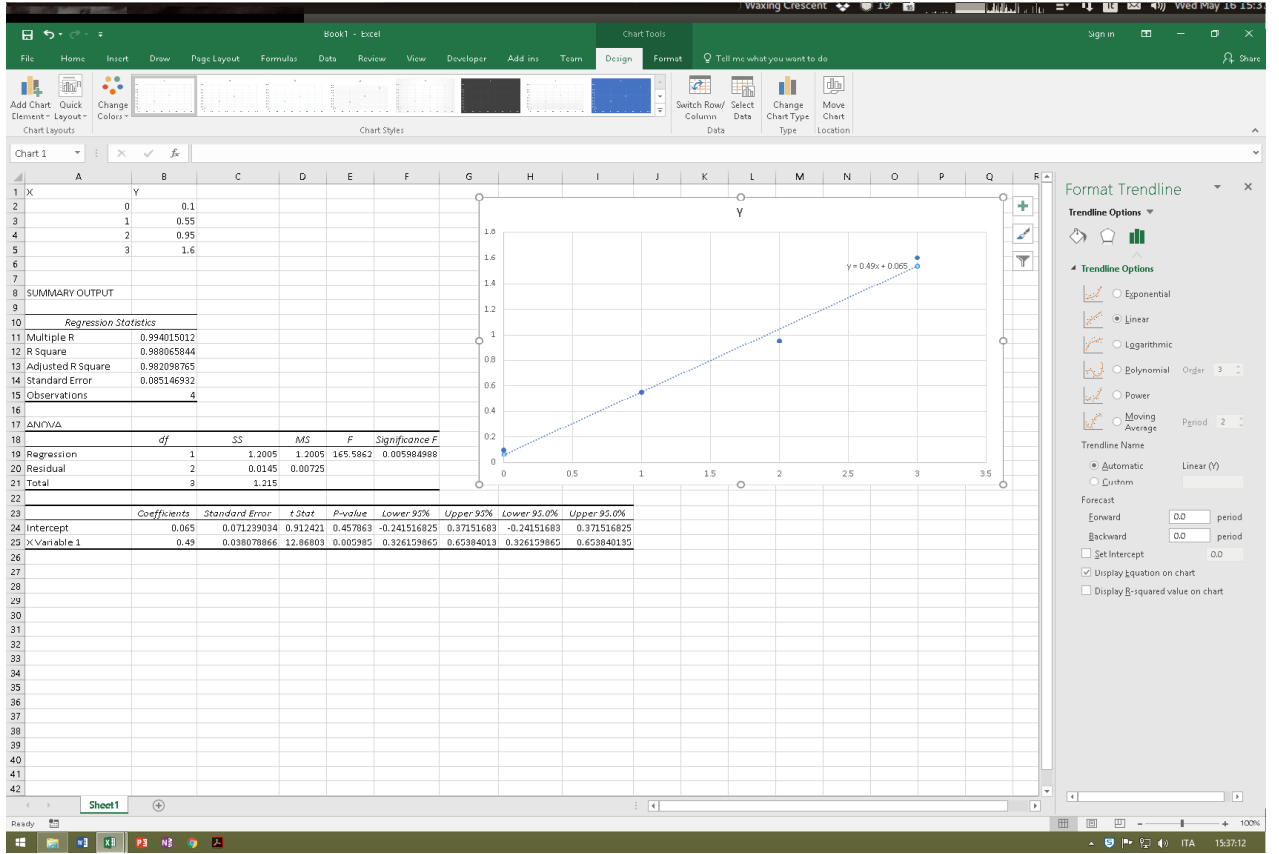


Figure 5: Least Squares technique in Microsoft Excel

The above expression is called the *Normal Equation* and gives as a solution:

$$\theta = (X^T X)^{-1} X^T y$$

where the superscript -1 defines the inverse of a matrix. So by constructing proper matrices and vectors and applying the above expression, the values for the parameters m and q that minimize the Mean Square Error ε_{MSE} can be easily obtained.

3.1.1 Example in Matlab/Octave

Let's apply the above formulation to the example proposed in Section 1.2, for which we already computed the wanted results in Section 2.2. Recall the 4 points were defined as:

$$P_1 = (0, 0.1) \quad P_2 = (1, 0.55) \quad P_3 = (2, 0.95) \quad P_4 = (3, 1.6)$$

So the matrices to be created are:

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad y = \begin{bmatrix} 0.1 \\ 0.55 \\ 0.95 \\ 1.6 \end{bmatrix}$$

The first column of X is all 1s, the second column contains the x_i coordinates. They can be entered directly with the following commands:

```
>> X = [1 0; 1 1; 1 2; 1 3]
>> y = [0.1; 0.55; 0.95; 1.6]
```

However, most often you may have a matrix P with the x_i and y_i coordinates, maybe loaded from a file, or entered with the command:

```
>> P = [0 0.1; 1 0.55; 2 0.95; 3 1.6]
```

In order to create the wanted matrix X and vector y you need to apply the following commands:

```
>> X = [ones(size(P, 1), 1) P(:, 1)]
>> y = P(:, 2)
```

Whatever method you use to enter the data, then the values of q and m that minimize the Mean Square Error ε_{MSE} can simply be obtained by:

```
>> theta = pinv(X' * X) * X' * y
theta =
    0.065000
    0.490000
```

where `pinv` is the pseudoinverse of a matrix, and corresponds to the inverse when the latter exists. The above results give:

$$\theta = \begin{bmatrix} q \\ m \end{bmatrix} = \begin{bmatrix} 0.065 \\ 0.49 \end{bmatrix}$$

which is exactly what was already obtained previously with the other formulation. Moreover, the minimum Mean Square Error $\varepsilon_{\text{MSE}}^{\min}$ is:

```
>> epsilonMSE = 1/size(y) * (y - X * theta)' * (y - X * theta)
epsilonMSE = 3.6250e-03
```

and the errors ε_i for each point are given by:

```
>> epsilon = y - X * theta
epsilon =
    3.5000e-02
   -5.0000e-03
   -9.5000e-02
    6.5000e-02
```

The matrix formulation can also be applied to the problem of *Multivariate Regression*, where there are multiple independent variables, with respect to this case where there is a single independent variable (i.e. the x_i coordinates of the points).